12th hacker series

Semiconductor

Electronics

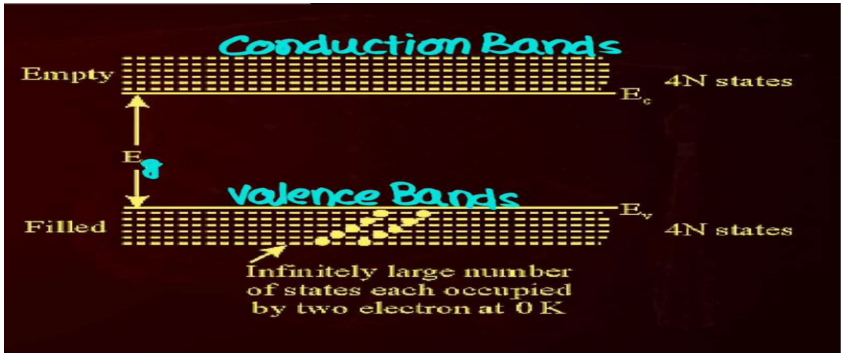
- a device whose functioning is based on control movement of electrons through it is called an electronic device
- Some of the present demo common such devices include a semiconductor junction diode, transistor, and integrated circuit
- The related branch in which we study the functioning and use of such devices is called electronics

Energy bands in solids

- an isolated atom has well defined energy levels. However, when large number of atoms get together to form a solid, these individual energy levels overlap and get completely modified.
- Inside the crystal, each electron has a unique position, and electrons have exactly same pattern of surrounding charges. Because of this, each electron will have a different energy level.
- Instead of discrete value of energy of electrons, the energy values lie in a certain range. The collection of these closely packed energy levels are set to form an energy band.

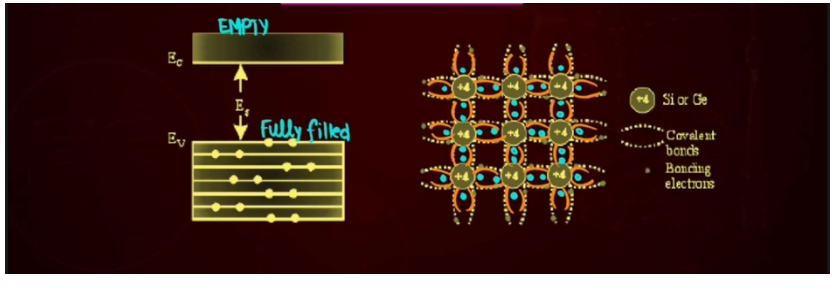
Valence band and conduction band

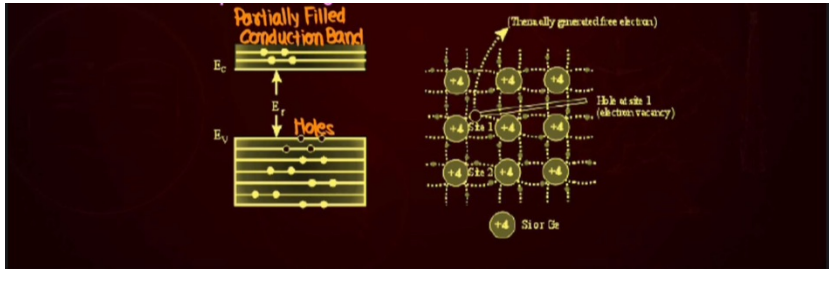
- two types of such bands formed in solids are called valence band and conduction band. The band formed by energy levels is known as valence band, whereas partially filled or unfilled band is known as conduction band.
- The two bands are generally separated by a gap called energy gap of forbidden gap. Depending upon the size of the energy gap, different materials behave as conductor, semiconductors, or insulator.
- The insulators have generally large energy gaps, whereas the conductors do not have any such gap. Semiconductors as small energy gap.



Formation of hole

- at zero Kelvin, no electron jumps from valence band to conduction band. It means germanium or silicon semiconductor at zero Kelvin behaves as an insulator.
- When the temperature of germanium or silicon semiconductor increases, then some electrons gain thermal energy due to thermal agitation and jump the conduction band from valence band, leaving behind equal number of holes in valence band as shown. This deficiency of electron is called hole. The hole behaves as a apparent free particle with effective positive charge.





Types of semiconductors- intrinsic and extrinsic

- common semiconductors are of two type- intrinsic and extrinsic. Germanium and Sillicon are two most commonly used semi semiconductor material.

Intrinsic semiconductor

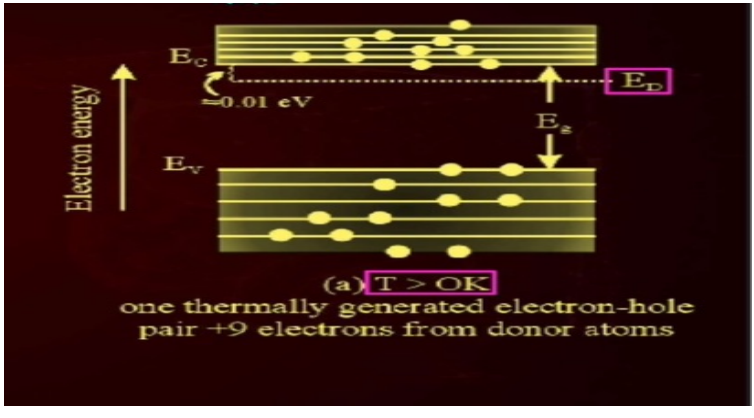
- pure semiconductors is in which the conductivity is caused due to charge carriers made available from within the material are called intrinsic semiconductors
- There are no free recharge carriers available under normal conditions. However, when the temperature is raise slightly, some of the covalent bonds in the material get broken due to thermal education and few electrons become free
- in order to fill the vacancy, created by absence of electron at a particular location electron from other position, move to this location and create a vacancy at another place called hole
- The movement of electrons and holes within the material result in conduction
- And intrinsic semiconductor, behave as a perfect insulator at temperature, zero Kelvin

Extrinsic semiconductors

- the semiconductors in which the conductivity is caused due to charge carriers, made available from external source by adding impurity from outside are called extrinsic semiconductor
- The process of adding impurity is called doping. The impurity added is generally from third group or fifth group. There are two types of dopant used in doping the tetravalent, silicon or germanium.
- 1) pentavalent- like arsenic, antimony, and phosphorus
- 2) trivalent- like Indium, boron, aluminium
- There are two types of extrinsic semiconductors
- 1) n- type semiconductor
- 2) p- type semiconductor
- if. n_i Is the density of intrinsic charge carriers n_e And n_h a density of electrons and whole intrinsic semiconductors, then the selection among them is $n_e n_h = n_i^2$

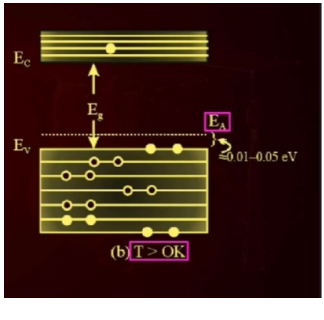
n- type semiconductor

- when a pentavalent impurity like phosphorus, antimony arsenic is doped in pure germanium, then the conductivity of Crystal increases due to surplus electrons and such a crystal is said to be n- type semiconductor while the impurity atoms are called donor atoms
- Thus in, n- type semiconductor, the charge carriers are negatively charged electrons and the donor level lies near the bottom of the conduction band



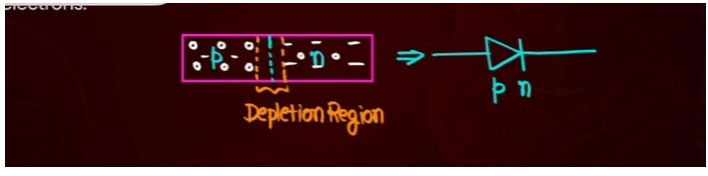
p- type semiconductor

- when a trivalent impurity like aluminium, Indian boron, gallium, etc is doped in pure germanium or silicon, then the conductivity of the crystal increases due to deficiency of electrons I.e, holes, and such a crystal is said to be p- type, semiconductor, while the impurity atoms are called acceptor
- Thus in, p- type semiconductor, the charge carriers are holes. Except a level lies near the top of the valence band.



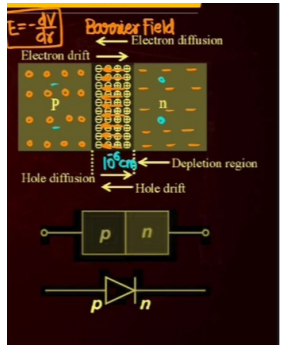
Semiconductor diode- p- n junction diode

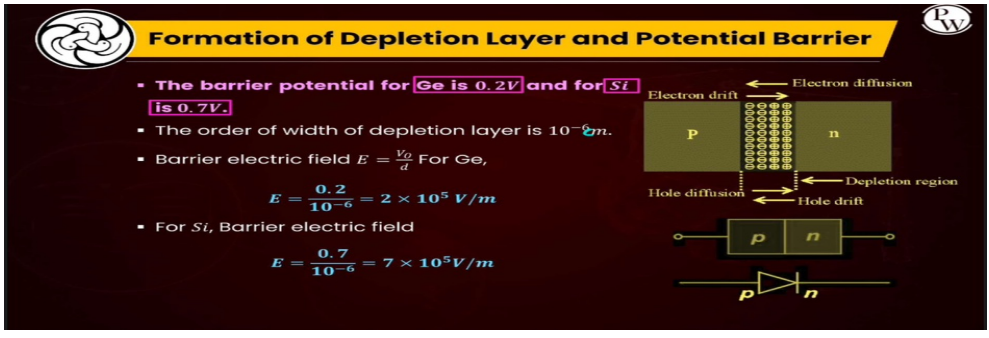
- a semiconductor having p- type impurity at one end and n - type impurity at the other end is known as p- n junction diode. The junction at which p- type and n- type, semiconductor combine is called p- n Junction
- in p- type region, there is majority of holes and in n- type region, there is majority of electrons



Formation of depletion layer and potential barrier

- at the Junction, there is diffusion of charge carriers due to thermal agitation. Therefore, some of the electrons of n - region diffuse to p- region, while some of the holes of p- region, diffuse into n - region
- Some charge carriers combine with opposite charges to neutralise each other. Does near the junction there is an axis of positively charged ions in n- region and an excess of negatively charged iron in p- region
- This set up a potential difference called potential barrier and hence an internal electric field across the junction. The potential barrier is usually of the order of μV.
- The field. E Is directed from n- region to p- region. This field stops the further diffusion of charge carriers. Thus, the layers on either side of the junction becomes free from mobile charge carriers and hence is called the depletion layer.

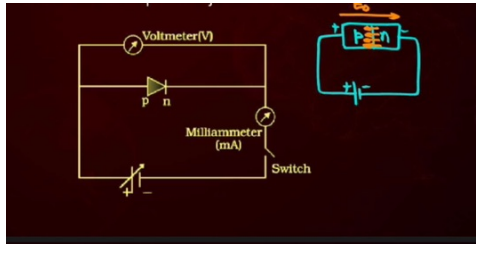




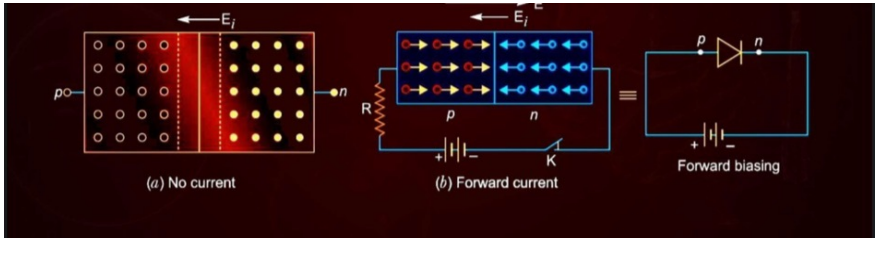
Forward bias

- in this arrangement, the positive terminal of battery is connected to p- end and negative terminal to n- and of the crystal so that an external electric field E established directed from p to n end to oppose the internal field, E.Thus, the junction is set to conduct

Under this arrangement, the whole move along the field E from p- region to n - region and electrons move opposite to field E from n- region to p- region, eliminating the depletion layer. Current is thus set up in the junction diode.

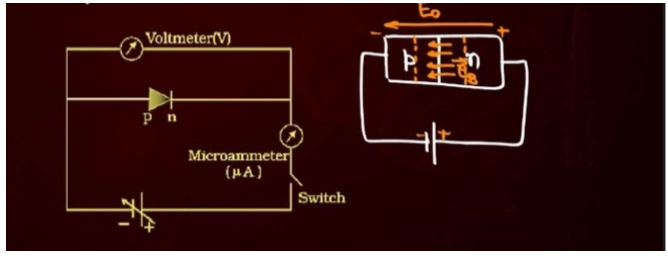


- within the junction diode, the current is due to both types of majority charge carriers, but in external circuit, it is due to electrons only
- The current is due to diffusion of majority charge carriers through the junction and is of the order of Milliamperes

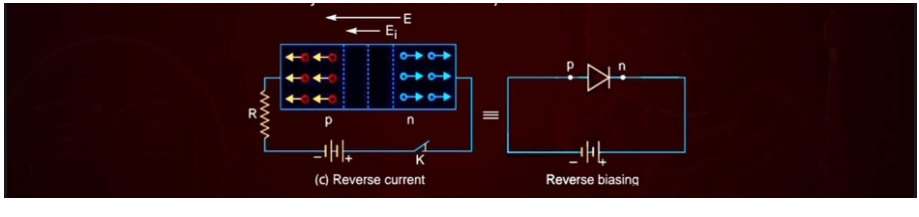


Reverse bias

- in this arrangement, positive terminal of battery is connected to n- end and and negative terminal to p- end of the crystal so that the external field established to support the internal field E_i as shown
- Under the biasing, the holes in p- region and the electrons in n- region are pushed away from the junction to hide the depletion layer and hence increases the size of the potential barrier. Therefore, the junction does not conduct.



- in the potential difference across the junction is increased in steps, a very small reverse current of the order to micro ampere flow
- The reason is that due to thermal agitation, some covalent bonds of pure semiconductor break, releasing a few holes in n- region and a few electrons in p- region called the minority charge carriers
- The reverse bias opposes the major majority charge carrier, but aids the minority charge carriers to move across the junction. Hence, a very small current flows.

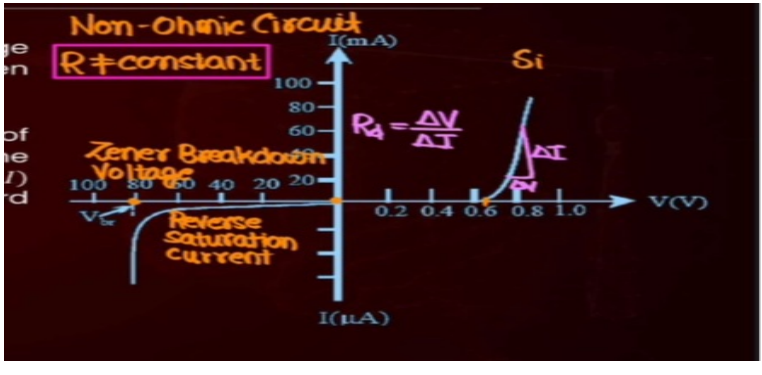


- the basic features of reverse bias are:
- 1) within the junction diode. The current is due to both types of minority charge carriers, but in external circuit, it is due to electrons only.
- 2) the current is due to leakage of minority charge carriers through the junction and is very small of the order of μA.

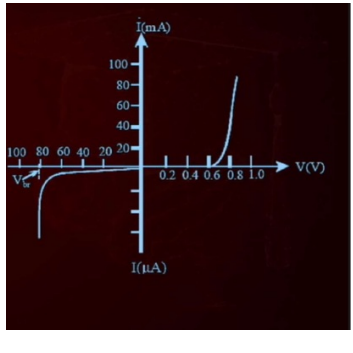
Characteristic of p – n junction diode

- breakdown voltage: it is the reverse voltage at which p- n junction breaks down with sudden rise in reverse current
- Dynamic or AC resistance: the ratio of change in forward voltage across the diode to the resulting change in current through. It is called dynamic or AC forward resistance of the diode.

$$r_d = \frac{\Delta V}{\Delta I}$$

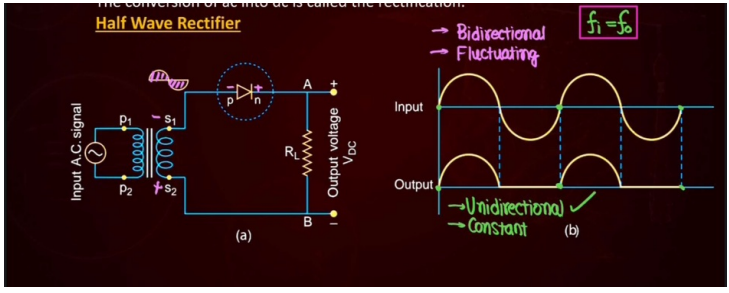


- avalanche breakdown: if the rivers wires is made sufficiently high, the covalent bond near the junction breakdown, releasing free electrons and holes
- These electrons and holes gain sufficient energy to break other covalent bond
- Does a large number of electrons in holes get free
- The reverse current increases abruptly to high value
- This is called avalanche breakdown and may damage the junction



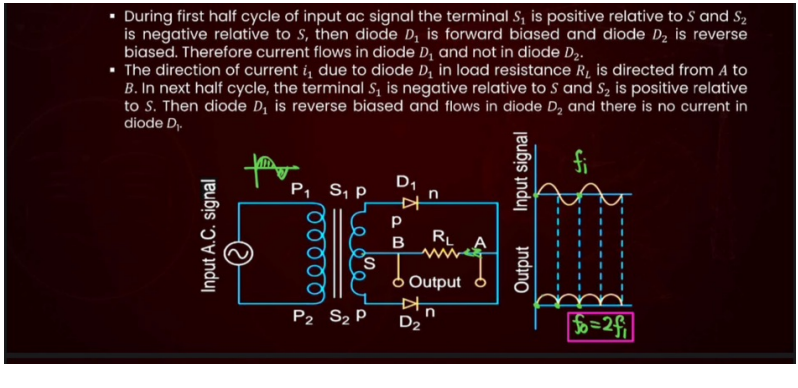
p – n junction diode as a half wave rectifier

- the conversion of AC into DC is called the rectification



- during first half of the input cycle, the secondary terminal or the transformer be positive relative to S2. Then the junction diode is forward biased. Therefore the current flows and its direction of current in lower resistance is from A to B
- in next half cycle, the terminal S one becomes negative relative to S2. Then the diode is in reverse by us. Therefore no current flows in diode and hence there is no potential difference across load R. The cycle repeats
- The output current in load flows only when S1 is positive relative to S2 that is during first half cycles of the input AC signal. There is current in circuit enhance potential difference across the resistance R while no current flows for the next half of this cycle.
- T the direction of current in Load is always from A to B which is direct current. Thus, a single p n junction diode acts as a half a rectifier.

p- n junction diode as a full wave rectifier



- The direction of current i_2 due to diode D_2 in load resistance is again from A to B Thus, for input ac signal the output current is a continuous series of unidirectional pulses.
- The input and output sequels are shown in the figure. This output current can be converted into steady current by the use of suitable filters.
- **In full wave rectifier if the fundamental frequency of input ac signal is 50 Hz, then the fundamental frequency of output is 100 Hz.**